Rules for differentiation

1

a $\dfrac{dy}{dx} = 6x$ b $\dfrac{dy}{dx} = 24x^3$ c $\dfrac{dy}{dx} = -12x^2$ d $\dfrac{dy}{dx} = 25x^4$

e $\dfrac{dy}{dx} = -\dfrac{3}{5}x^2$ f $\dfrac{dy}{dx} = 9x^5$ g $\dfrac{dy}{dx} = \dfrac{5x^3}{3}$ h $\dfrac{dy}{dx} = 4x^{-0.5}$

2

a $\dfrac{dy}{dx} = \dfrac{56}{9}x^{-\frac{2}{9}}$ b $\dfrac{dy}{dx} = 1.8x^{-0.4}$ c $\dfrac{dy}{dx} = 10x^{-6}$ d $\dfrac{dy}{dx} = -21x^{-4}$

3

a $\dfrac{dy}{dx} = \dfrac{4}{\sqrt{x}}$ b $\dfrac{dy}{dx} = \dfrac{1}{8\sqrt{x}}$ c $\dfrac{dy}{dx} = 35\sqrt{x^3}$ d $\dfrac{dy}{dx} = \dfrac{-1}{\sqrt{x^3}}$

4

a $y = 7x^{-1}$ b $\dfrac{dy}{dx} = -\dfrac{7}{x^2}$

5

a $\dfrac{dy}{dx} = -\dfrac{21}{x^{\frac{5}{2}}}$ b $\dfrac{dy}{dx} = -\dfrac{3}{4x^4}$

6

a $\dfrac{dy}{dx} = 1$ b $\dfrac{dy}{dx} = 2$

c $\dfrac{dy}{dx} = \dfrac{2}{9}$ d $\dfrac{dy}{dx} = 2x - 1$

e $\dfrac{dy}{dx} = 2x - 8$ f $\dfrac{dy}{dx} = 4x^3 + 5x^4$

g $\dfrac{dy}{dx} = 5x^4 - 4x^3$ h $\dfrac{dy}{dx} = 6x^2 - 6x - 4$

i $\dfrac{dy}{dx} = \dfrac{5x^4}{2} + \dfrac{8x^7}{5}$ j $\dfrac{dy}{dx} = -15x^4 + 20x^3 - 15x^2 - 8x + 2$

k $\dfrac{dy}{dx} = x^7 + x^4 - 3$ l $\dfrac{dy}{dx} = -18x^{-7} - \dfrac{3x^{-4}}{7}$

m $\dfrac{dy}{dx} = -\dfrac{120}{x^6} + \dfrac{120}{x^5}$ n $\dfrac{dy}{dx} = \dfrac{x^{-\frac{1}{2}}}{2} + 6x^{-\frac{1}{4}}$

7 $\dfrac{dy}{dx} = 49ax^6 - 6bx^2$

8

a $\dfrac{dy}{dx} = 3x^2 - \dfrac{5}{x^6}$ b $\dfrac{dy}{dx} = \dfrac{1}{3x^{\frac{2}{3}}} - \dfrac{7}{x^8}$

9

a $f(r) = 2r^{-1} + \frac{r}{3}$

b $f'(r) = -2r^{-2} + \frac{1}{3}$

10 $\frac{dy}{dx} = -\frac{2a}{x^{a+1}} + \frac{3b}{x^{b+1}}$

11

a

i $y = -\frac{16x}{9} - \frac{32}{9}$

ii $\frac{dy}{dx} = -\frac{16}{9}$

b

i $y = 6x^2 + 23x + 15$

ii $\frac{dy}{dx} = 12x + 23$

c

i $y = 14x^3 + 4x^2$

ii $\frac{dy}{dx} = 42x^2 + 8x$

d

i $y = 15x^3 + 38x^2 + 24x$

ii $\frac{dy}{dx} = 45x^2 + 76x + 24$

e

i $y = x^2 + 8x + 16$

ii $\frac{dy}{dx} = 2x + 8$

f

i $y = 5x^2 - 30x + 45$

ii $\frac{dy}{dx} = 10x - 30$

g

i $y = 64x^2 - 64x + 16$

ii $\frac{dy}{dx} = 128x - 64$

h

i $y = 12x^{\frac{3}{2}} + 10x + 18x^{\frac{1}{2}} + 15x^{-1}$

ii $\frac{dy}{dx} = 18x^{\frac{1}{2}} + 10 - 9x^{-\frac{3}{2}} + 15x^{-2}$

12

a $\frac{dy}{dx} = 42x + 32$

b $\frac{dy}{dx} = 2x + 1$

13

a $f(x) = x + 20x^{\frac{5}{2}} + 100x^4$

b $f'(x) = 1 + 50\sqrt{x^3} + 400x^3$

14

a

i $f(x) = 4 + \frac{3}{2}x^{-1}$

ii $f'(x) = -\frac{3}{2}x^{-2}$

b

i $y = 4x^7 - 2x^6 + 3x^5 + \frac{9}{2}x^{-2}$

ii $\frac{dy}{dx} = 28x^6 - 12x^5 + 15x^4 - 9x^{-3}$

c

i $y = 3x^3 + 9x + x^{-\frac{5}{2}} + 11x^{-3}$

ii $\frac{dy}{dx} = 9x^2 + 9 - \frac{5}{2x^{\frac{7}{2}}} - 33x^{-4}$

d

i $y = 8x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + 4x^{-\frac{1}{2}}$

ii $\frac{dy}{dx} = 12x^{\frac{1}{2}} + 3x^{-\frac{1}{2}} - 2x^{-\frac{3}{2}}$

15

a $y = \frac{5}{4}x^{-\frac{7}{2}}$

b $\frac{dy}{dx} = -\frac{35}{8}x^{-\frac{9}{2}}$

16

a $\frac{dy}{dx} = 4x^3 - \frac{5}{x^6}$

b $\frac{dy}{dx} = -\frac{5}{x^6} + 6x^5$

c $\frac{dy}{dx} = \frac{6}{7x^{\frac{1}{7}}} - \frac{1}{7x^{\frac{8}{7}}}$

d $\frac{dy}{dx} = \frac{7x^{\frac{1}{6}}}{6} - \frac{10}{3x^{\frac{1}{3}}}$

Gradients

17 $f'(2) = 22$

18 $f'(4) = 0$

19 $f'(-4) = 2$

20 $f'(0) = \frac{1}{2}$

21

a 1

b $x = 1$

22

a $f'(2) = -3$

b $f'(2) = 39$

c $f'(25) = -\frac{3}{125}$

d $f'(4) = -\frac{1855}{4}$

e $f'(4) = -1$

23

a $f'(x) = 12x + 5$

b $f'(2) = 29$

c $x = 3$

24

a $f'(x) = 3x^2 - 4$ b $f'(4) = 44$ c $f'(-4) = 44$ d $x = \pm 5$

25

a $\frac{dy}{dx} = 4x - 8$ b $x = 2$

26 $x = \pm 3$

27 $x = \frac{1}{144}$